

Question 1: Class Student

Write a class Student with attributes name and age. Create an object and print its details.

```
class Student:
    def __init__(self, name, age):
        self.name = name
        self.age = age

# Create an object
student1 = Student("Alice", 20)

# Print its details
print(f"Student Name: {student1.name}, Age: {student1.age}")
```

Student Name: Alice, Age: 20

Question 2: Class Car

Create a class Car with attributes brand and model. Write a method display() to print the car details.

```
class Car:
    def __init__(self, brand, model):
        self.brand = brand
        self.model = model

    def display(self):
        print(f"Car Brand: {self.brand}, Model: {self.model}")

# Create an object and call display method
car1 = Car("Toyota", "Camry")
car1.display()
```

Car Brand: Toyota, Model: Camry

Question 3: Class Book

Write a class Book with attributes title and author. Create two objects and display their information.

```
class Book:
    def __init__(self, title, author):
        self.title = title
        self.author = author

# Create two objects
book1 = Book("The Hobbit", "J.R.R. Tolkien")
book2 = Book("Pride and Prejudice", "Jane Austen")

# Display their information
print(f"Book 1 - Title: {book1.title}, Author: {book1.author}")
print(f"Book 2 - Title: {book2.title}, Author: {book2.author}")
```

Book 1 - Title: The Hobbit, Author: J.R.R. Tolkien
Book 2 - Title: Pride and Prejudice, Author: Jane Austen

Question 4: Class Person

Create a class Person with a method greet() that prints "Hello, welcome!".

```
class Person:
    def greet(self):
        print("Hello, welcome!")

# Create an object and call greet method
person1 = Person()
person1.greet()
```

Hello, welcome!

Question 5: Class Rectangle

Write a class Rectangle with attributes length and width. Create a method area() to calculate the area.

```
class Rectangle:
    def __init__(self, length, width):
        self.length = length
        self.width = width

    def area(self):
        return self.length * self.width

# Create an object and calculate area
rectangle1 = Rectangle(5, 10)
print(f"Rectangle Area: {rectangle1.area()}")
```

Rectangle Area: 50

Question 6: Class Circle

Create a class Circle with an attribute radius. Write a method circumference() to calculate the circumference.

```
import math

class Circle:
    def __init__(self, radius):
        self.radius = radius

    def circumference(self):
        return 2 * math.pi * self.radius

# Create an object and calculate circumference
circle1 = Circle(7)
print(f"Circle Circumference: {circle1.circumference():.2f}")
```

Circle Circumference: 43.98

Question 7: Class Dog

Write a class Dog with attributes name and breed. Add a method bark() that prints "Woof!".

```
class Dog:
    def __init__(self, name, breed):
        self.name = name
        self.breed = breed

    def bark(self):
        print("Woof!")

# Create an object and call bark method
dog1 = Dog("Buddy", "Golden Retriever")
print(f"Dog Name: {dog1.name}, Breed: {dog1.breed}")
dog1.bark()
```

Dog Name: Buddy, Breed: Golden Retriever
Woof!

Question 8: Class Employee

Create a class Employee with attributes name and salary. Write a method to display the employee details.

```
class Employee:
    def __init__(self, name, salary):
        self.name = name
        self.salary = salary

    def display_details(self):
        print(f"Employee Name: {self.name}, Salary: ${self.salary:,}")

# Create an object and display details
employee1 = Employee("John Doe", 60000)
employee1.display_details()
```

Employee Name: John Doe, Salary: \$60,000

Question 9: Class Laptop

Write a class Laptop with attributes brand and price. Create an object and print both values.

```
class Laptop:
    def __init__(self, brand, price):
        self.brand = brand
        self.price = price

# Create an object
laptop1 = Laptop("Dell", 1200)

# Print both values
print(f"Laptop Brand: {laptop1.brand}, Price: ${laptop1.price:,}")
```

Laptop Brand: Dell, Price: \$1,200

Question 10: Class BankAccount

Create a class BankAccount with an attribute balance. Write a method show_balance() to display the balance.

```
class BankAccount:
    def __init__(self, balance):
        self.balance = balance

    def show_balance(self):
        print(f"Current Balance: ${self.balance:,.2f}")

# Create an object and display balance
account1 = BankAccount(1500.75)
account1.show_balance()
```

Current Balance: \$1,500.75